

Module4 Assignment

```
// Problem 1: Student class
class Student {
    String name;
    int roll;

    // Constructor
    Student(this.name, this.roll);

    // Method to display info
    void displayInfo() {
        print("Name: $name, Roll: $roll");
    }
}

void main() {
    // Create Student object
    Student student1 = Student("Rahim", 101);
    student1.displayInfo();

    // Problem 2: Person and Teacher classes
    Person person1 = Person("Karim");
    Teacher teacher1 = Teacher("Hasan", "Mathematics");
    teacher1.displayInfo();

    // Problem 3: Abstract class Shape and Circle subclass
    Circle circle1 = Circle(5.0);
    print("Circle Area: ${circle1.area}");
}

// Problem 2: Person and Teacher classes
class Person {
    String name;

    Person(this.name);
}

class Teacher extends Person {
    String subject;

    Teacher(String name, this.subject) : super(name);

    void displayInfo() {
```

```
    print("Name: $name, Subject: $subject");  
  }  
}
```

// Problem 3: Abstract class Shape and Circle subclass

```
abstract class Shape {  
    double _area = 0;  
  
    // Abstract method  
    void calculateArea();  
  
    // Getter  
    double get area => _area;  
  
    // Setter  
    set area(double value) {  
        _area = value;  
    }  
}
```

```
class Circle extends Shape {  
    double radius;  
  
    Circle(this.radius) {  
        calculateArea();  
    }  
  
    @override  
    void calculateArea() {  
        area = 3.1416 * radius * radius;  
    }  
}
```